

Zheng XU

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EDUCATION

University of Manchester

Sep. 2023 – Present

BEng (Hons) Electrical and Electronic Engineering

- **Results: First Class (Expected)**
- **Core modules:** Microcontroller Engineering I & II, Electronic Circuit Design I, Digital Systems Design I & II, Energy Transport and Conversion, Engineering Management, etc.

RESEARCH PROJECTS

BCI-based Robotic Arm Control System (Deep Reinforcement Learning)

Sep. 2025 - Present

University of Manchester Independent Project

- Led the end-to-end development of an intelligent control framework that translates non-stationary EEG signals from Brain–Computer Interfaces into precise physical robotic actuation using deep reinforcement learning.
- Engineered a Transformer-based sequence controller integrated with a Deep Q-Network (DQN) framework in PyTorch; leveraged temporal context for movement intent prediction, significantly stabilizing trajectory smoothness and reducing convergence steps by 34% compared to standard memoryless agents.
- Optimized algorithmic convergence by designing an adaptive step-size strategy that dynamically adjusts based on target distance, effectively accelerating the spatial targeting phase by 47% without compromising precision.
- Addressed cross-subject variability by processing and integrating the large-scale 64-channel PhysioNet dataset alongside BCI IV-2a/2b using MNE-Python; trained joint models across 20+ subjects utilizing early-stopping regularization to successfully mitigate overfitting.
- Successfully deployed the optimized AI agent from simulation to a physical SO-101 robotic arm, resolving real-world mechanical constraints by implementing synchronized multi-joint control and dynamic velocity compensation for loaded joints.

Hai'an Institute of Intelligent Equipment, Shanghai Jiao Tong University

Jun. 2025 - Jul. 2025

Research Intern

- Conducted research and development on robotic arm control systems, focusing on motion accuracy and dynamic response optimisation.
- Designed and implemented control algorithms in MATLAB and embedded C++ for trajectory planning and feedback regulation.
- Prepared experimental reports and presented findings to the laboratory's research supervisors.

Embedded Systems Project

Sep. 2024 - Apr. 2025

- Designed and implemented an autonomous line-following robot using an STM32 Nucleo-F401RE microcontroller.
- Integrated and characterised multiple sensors — TCRT5000 IR, MPU-6050 gyroscope, and HC-SR04 ultrasonic — for real-time environmental perception.
- Developed a PID control algorithm in C++ for accurate path-tracking and obstacle avoidance.
- Designed and tested custom PCBs in Altium Designer; performed sensor calibration and signal conditioning to reduce noise.
- Conducted sensor characterisation experiments (ambient-light response, line spread function, voltage–distance curve) to optimise performance and stability.
- Implemented Bluetooth Low Energy (BLE) communication for wireless data logging and remote debugging of the embedded system.

Quanser Aero 2 Control Systems Project

Jun. 2023 - Jul. 2023

- Modelled and analysed the Quanser Aero 2 dual-rotor platform using MATLAB and Simulink.

- Identified rotor and pitch subsystem dynamics through experimental data; estimated system parameters such as gain, damping ratio, and natural frequency.
- Designed and tuned P, PI, and PID controllers to improve transient and steady-state performance.
- Compared simulated and experimental responses, investigating deviations caused by unmodelled nonlinearities, actuator limitations, and frictional effects.
- Demonstrated proficiency in system identification, control theory, and real-time hardware-in-the-loop testing using QUARC.

EXTRACURRICULAR ACTIVITIES

- **American Football Club Member:** Active participant in the University of Manchester's American Football team, developing teamwork, discipline, and strategic thinking.
- **Crystal Growing Hobbyist:** Cultivated over 15 distinct types of crystals and presented results at local science fairs, demonstrating experimental precision and long-term project commitment.

SKILLS & HOBBIES

- **Technical Skills:** Microsoft Office Suite; Sentaurus TCAD (device simulation); Origin (Data visualisation and graph plotting)
- **Language Skills:** Chinese (native), English (proficient)
- **Programming Languages:** C, C++, Assembly, Python, MATLAB, VHDL
- **Microcontroller Programming:** Assembly for PIC series; C++ for STM32
- **Digital Systems Design:** VHDL, Digital Signal Processing (DSP)
- **Simulation & Modelling:** MATLAB, Simulink, LabVIEW, SolidWorks, Altium Designer
- **Circuit Design:** PCB design, signal analysis, oscilloscope and multimeter operation
- **Data Analysis:** MATLAB Data Analyzer, Excel, Python (Pandas)
- **Embedded Systems:** Real-time control, sensor integration, and hardware debugging
- Certified MATLAB Associate -- *MathWorks*
- Certified SolidWorks Professional (CSWP) -- *Dassault Systèmes*
- Python Programming for Engineers -- *Coursera*
- **Interest:** Anime, Computer game